

0DTE 15:30–16:00 nearest-OTM package: per-rule return summary, models compared on the same 871 days

	n	mean	std	min	25%	50%	75%	max	skew	ex. kurt	t	Sharpe _{ann}	n_{buy}	%buy
<i>Short volatility (always short)</i>														
all models (no forecast)	871	0.020	1.127	-10.32	-0.444	0.329	0.931	1.00	-2.31	10.90	0.53	0.28	0	0.0
<i>sign(s) (± 1 on the sign of s)</i>														
baseline (HAR + calendar OLS)	871	0.071	1.125	-5.42	-0.635	0.176	0.883	10.32	0.54	10.68	1.85	1.00	280	32.1
block-diag ridge	871	0.115	1.121	-5.42	-0.686	0.151	0.912	10.32	0.96	10.56	3.03	1.63	339	38.9
LightGBM	871	0.094	1.123	-5.42	-0.639	0.176	0.900	10.32	0.90	10.58	2.48	1.33	282	32.4
XGBoost	871	0.098	1.123	-5.42	-0.624	0.180	0.914	10.32	0.84	10.60	2.58	1.39	275	31.6
lasso (causally tuned)	871	0.102	1.122	-6.84	-0.650	0.172	0.914	10.32	0.44	10.75	2.68	1.44	320	36.7
lasso ($\alpha = 10^{-4}$)	871	0.130	1.120	-5.42	-0.698	0.124	0.892	10.32	1.29	10.42	3.42	1.84	365	41.9
elastic net (causally tuned)	871	0.080	1.124	-6.84	-0.660	0.147	0.913	10.32	0.44	10.71	2.11	1.14	303	34.8

One expiration day = one return; mid fill. $t = \sqrt{n} \cdot \text{mean}/\text{std}$. “ex. kurt” is Fisher excess kurtosis (fourth standardized moment minus 3; Gaussian = 0). $n_{\text{buy}}/\% \text{buy}$ count days the rule is net long ($q_t > 0$). The short-volatility rule takes no forecast, so it is one row for all models. Sharpe_{ann} = (mean/std) $\times \sqrt{252}$; every other column is daily.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties); LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned).